

ELECTROMAGNETIC COMPATIBILITY (EMC) –

Part 3-12: Limits – Limits for harmonic currents produced by equipment connected to public low-voltage systems with input current >16 A and ≤ 75 A per phase

INTERPRETATION SHEET

This interpretation sheet has been prepared by subcommittee 77A: Low frequency phenomena, of IEC technical committee 77: Electromagnetic compatibility.

The text of this interpretation sheet is based on the following documents:

ISH	Report on voting
77A/792/ISH	77A/800/RVD

Full information on the voting for the approval of this interpretation sheet can be found in the report on voting indicated in the above table.

Interpretation of requirements for equipment with unforeseen low input currents during tests according to IEC 61000-3-12:2011: Electromagnetic compatibility (EMC) – Part 3-12: Limits – Limits for harmonic currents produced by equipment connected to public low-voltage systems with input current >16 A and ≤ 75 A per phase.

When equipment that has a rated current above 16 A draws a reference current that is less than 16 A under the specified test conditions, the manufacturer may proceed in one of the following ways:

- 1) Comply with the proportional limits as calculated, choosing the required R_{sce} ;
- 2) Comply with the absolute limits given in IEC 61000-3-2:2011, Table 1, using the measurement procedure defined in 4.2.2 in IEC 61000-3-12:2011;
In that case, the manufacturer shall state in the instruction manual “Equipment complying with IEC 61000-3-12”, without having to declare a minimum short circuit power S_{sc} .
- 3) Change the test conditions to a representative 2,5 min period, as defined in IEC 61000-3-12:2011, Table 1 for long cyclic equipment, and comply with the proportional limits as calculated, choosing the required R_{sce} .

The manufacturer is strongly advised to state in the test report which of these ways was used, so that subsequent tests are carried out with the same procedure.

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